

THE INTERNET OF THINGS



News, views and development on the Internet of Things

“Bringing the Internet of Things to life requires a comprehensive systems approach...”

Narang N. Kishor, mentor and principal design architect of Namix Technolabs helps us cut through the jargon and look deeper into what really comprises the IoT

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The IoT is probably too broad a term for people to focus on. Could you, specifically, name some of the exciting technologies that are beginning to surround us today (signs of coming of the IoT)?

Well, IoT is all about heterogeneous and aware devices interacting to simplify people's life in some way or the other.

One of the most profound (exciting as well as terrifying) technologies is augmented reality (AR). It is altering the way the world or immediate environment is viewed, especially for the users of this technology. Imagining the way the world would appear and be experienced, while walking or driving down the street.

With AR displays, informative graphics will appear in the corner of the glass, and an audio prompt would inform about whatever one is seeing. This information will be refreshed in tandem with movements of the wearer. Similar technology is already available in smartphones. And Google Glasses are already here.

Another up-and-coming technology is the ability of smartphones to communicate via NFC and Wi-Fi with a range of devices, including wrist-watches, healthcare sensors and home

entertainment systems. People are already captivated by the vision of being able to control everything in their homes and offices, from temperature, lighting and security to using devices to brew cups of coffee, program entertainment, check health records and conduct a myriad of other tasks.

Enterprises are also beginning to embrace IoT for tracking physical assets, managing customer relationships and creating efficiencies in business operations and supply chains.

What do you think are the key components of the IoT—and at what stages of maturity are these? Which of these components/frameworks do you think need to really develop very fast in order to give the IoT the required momentum?

The key components of the IoT are sensing nodes, local embedded processing nodes, connectivity nodes, software to automate tasks and enable new classes of services, remote embedded processing nodes and last but not the least full security across the signal path.

To give the IoT the required momentum, device management platforms, cloud computing and big data sciences in the virtual world are already flourishing and well geared up to meet the expectations of the design-

ers. But, the physical world needs to catch up. The wireless and networking technologies need to mature very fast; we need innovative approaches in deployment of large sensor (and actuator) networks.

Bringing the IoT to life requires a comprehensive systems approach, inclusive of intelligent processing and sensing technology, connectivity, software and services, along with a leading ecosystem of partners.

For people/start-ups wishing to work in this field, what kind of competencies do they need to develop?

I would sincerely recommend them to brush up their basic electronics fundamentals including but not limited to sensors, analogue, power, RF and various compliance issues in hardware design. It would really help if they have in-depth understanding of different communication technologies, protocols and standards, particularly relevant to the domains of their applications. Understanding of security, privacy and socio-ethical implications of the solutions, services and applications they are trying to develop would also help. Please note, embedded processing, data analytics and other software skills are already considered as a must, so I am not delving into those.